SMS SPAM DETECTION USING DYNAMIC EVIDENCE ROUTING:
A TRANSFORMER-BASED APPROACH

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DECLARATION

We hereby declare that the mini-project report entitled "SMS SPAM DETECTION USING DYNAMIC EVIDENCE ROUTING: A TRANSFORMER-BASED APPROACH" is a record of bona fide work carried out by U. Jaya Sindhura (2420030044), K. Ushasri (2420030376), and P. Smitha (2420030583), in partial fulfillment of the requirements for the award of the Bachelor of Technology degree at K L (Deemed to be) University. The work presented in this report is original to the best of our knowledge. All datasets, tools, code, and external sources used have been appropriately acknowledged.

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CERTIFICATE

This is to certify that the mini-project report entitled “SMS SPAM DETECTION USING DYNAMIC EVIDENCE ROUTING: A TRANSFORMER-BASED APPROACH” is a bona fide work carried out by U. Jaya Sindhura (2420030044), K. Ushasri (2420030376), and P. Smitha (2420030583), under the supervision of Dr. K. Swanthana, during the academic year 2025–2026, in partial fulfillment of the requirements for the award of the Bachelor of Technology degree at K L (Deemed to be) University.

Signature of the Project Supervisor

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Signature of the HOD **Signature of the External Examiner**

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ABSTRACT

SMS communication remains an important channel for personal communication, commercial notifications, service updates, and organizational interactions. However, it is also used to distribute unsolicited advertisements, fraudulent offers, deceptive notifications, and messages intended to obtain sensitive information. Detecting SMS spam is challenging because messages are short, informal, and highly variable in vocabulary and structure.

This paper presents an SMS spam detection approach based on a Dynamic Evidence Routing Transformer. The implemented system uses a DistilBERT-based representation and combines three evidence pathways: semantic evidence, pattern evidence, and context/entity evidence. A routing mechanism assigns relative contributions to these pathways before the final HAM or SPAM classification. The system also exposes prediction confidence and routing values for analysis.

The reported evaluation was conducted on a test set containing 517 SMS messages. The proposed model achieved 99.42% accuracy, 100.00% precision, 95.38% recall, and an F1-score of 97.64%. Its confusion matrix contained 452 true negatives, 62 true positives, three false negatives, and zero false positives. The proposed model was also compared with TF-IDF-based Logistic Regression and Linear Support Vector Machine baselines. An ablation study examined the effects of removing individual evidence pathways and replacing dynamic routing with static equal fusion.

The results show strong classification performance under the reported evaluation conditions. However, static equal fusion and removal of the context/entity pathway produced the same aggregate metrics as the complete model. Therefore, the experiments do not establish that dynamic routing consistently improves classification over fixed fusion. Further evaluation on additional datasets, repeated runs, and larger robustness tests is required.

Keywords— SMS spam detection, text classification, natural language processing, Transformer models, DistilBERT, dynamic evidence routing, semantic analysis, pattern detection, machine learning.

1. INTRODUCTION

1.1 Background

Short Message Service (SMS) is widely used because it is simple, accessible, and supported by almost all mobile communication systems. It allows individuals to exchange messages without requiring advanced applications or continuous internet connectivity. Businesses and service providers also use SMS to communicate transaction updates, delivery notifications, appointment reminders, promotional offers, and account-related information.

Despite these advantages, SMS is also a channel for unwanted and deceptive communication. Spam messages may advertise products without consent, promote fraudulent prizes, request personal information, or persuade recipients to visit suspicious websites. Some messages imitate legitimate organizations to create a false impression of trust. These activities can reduce user confidence in mobile communication and expose recipients to financial or privacy risks.

The growing diversity of message content makes spam detection a continuing classification problem. Messages differ in length, writing style, vocabulary, punctuation, abbreviations, and intended purpose. A legitimate message may contain a promotional word, telephone number, or reference to a financial transaction. Similarly, a spam message may avoid familiar spam keywords and use ordinary conversational language.

An automated SMS spam detection system attempts to distinguish legitimate messages, commonly referred to as ham, from spam messages. Such a system must learn useful differences between the classes from labelled examples while limiting both false-positive and false-negative predictions. The importance of content-based SMS filtering and the distinctive characteristics of SMS data have been discussed in earlier work [1], [2].

1.2 Evolution of Spam Detection Techniques

Early spam-filtering systems often relied on manually defined rules and keyword matching. These systems searched for known suspicious expressions, repeated promotional phrases, unusual formatting, or predefined indicators. Their main advantage was simplicity: individual rules could be inspected and understood easily. However, rule-based systems depend on the quality and coverage of their rules. New spam messages can use unfamiliar wording, while legitimate messages can accidentally satisfy a rule.

Machine-learning techniques introduced a more flexible approach. Instead of specifying every classification rule manually, a model learns relationships between text features and message labels from a labelled dataset. Statistical representations such as term frequency–inverse document frequency (TF-IDF) can be combined with classifiers such as Logistic Regression and Support Vector Machines [3]–[6].

1. INTRODUCTION (CONTINUED)

1.3 Motivation

SMS messages can provide different types of evidence about their likely class. Some contain recognizable patterns, such as repeated promotional expressions or a direct instruction to claim an unexpected reward. Other messages use less obvious wording, where the relationship between several expressions is more informative than a single keyword. Contextual or entity-related indicators may also contribute information.

A model that treats all messages as though they depend on the same type of evidence may not use every useful signal effectively. A flexible architecture can combine different representations and adjust their relative contributions according to the input. Transformer-based language models provide contextual representations, while separate pathways can be used to process different forms of evidence [7]–[10].

The proposed Dynamic Evidence Routing Transformer is motivated by this need for input-dependent evidence combination. It uses three pathways: semantic evidence, pattern evidence, and context/entity evidence. The routing mechanism combines their outputs before the final classification. The study examines the classification performance of the implemented system and analyzes its routing behaviour.

1.4 Problem Statement

The primary problem addressed in this research is the accurate classification of SMS messages when useful indicators of spam vary from one message to another. SMS messages are usually short and may contain limited contextual information. Spam senders can change vocabulary, punctuation, and message structure. Legitimate messages can share individual words or expressions with spam messages. Some deceptive messages depend on meaning and context rather than obvious keywords.

The study investigates whether combining semantic, pattern, and context/entity evidence through dynamic routing can provide an effective solution for binary SMS spam classification. The evaluation compares the implemented model with conventional TF-IDF-based baselines and examines the effect of removing individual pathways.

1.5 Research Objectives

- Implement an SMS classification model using a DistilBERT-based text representation.
- Incorporate semantic, pattern, and context/entity evidence pathways within the classification architecture.
- Combine evidence pathways through the implemented routing mechanism.
- Classify messages as ham or spam and report prediction-confidence information.
- Compare the proposed model with TF-IDF-based Logistic Regression and Linear SVM baselines.
- Evaluate performance using accuracy, precision, recall, F1-score, and a confusion matrix.
- Investigate the effects of removing individual evidence pathways and replacing dynamic routing with static equal fusion.

1. INTRODUCTION (CONTINUED)

1.6 Research Questions

- RQ1: How effectively does the Dynamic Evidence Routing Transformer classify SMS messages on the reported test set?
- RQ2: How does its performance compare with TF-IDF-based Logistic Regression and Linear SVM when evaluated on the same reported split?
- RQ3: How does removing an individual evidence pathway affect the classification metrics?
- RQ4: How does the model distribute routing weights across the three evidence pathways?
- RQ5: What additional validation is required to determine whether the model can generalize to different message sources?

1.7 Contributions

The work makes an implementation-oriented contribution by bringing together a DistilBERT-based text representation, three evidence-processing pathways, and a routing-based fusion mechanism in an SMS classification system. It evaluates the resulting model against two conventional TF-IDF baselines using the reported test set. It also reports an ablation analysis and average and example-level routing distributions.

The contribution should be interpreted within the scope of the experiments. The paper does not claim that the architecture is the first use of multiple pathways or dynamic fusion in machine learning. It documents a specific model configuration and its observed behaviour on the evaluated SMS data.

1.8 Organization of the Paper

The remainder of the paper is organized as follows. Section 2 reviews existing SMS spam detection approaches and relevant text-representation methods. Section 3 describes the proposed methodology, dataset representation, preprocessing, model components, and algorithm. Section 4 presents the experimental setup and evaluation protocol. Section 5 reports classification results, baseline comparisons, confusion-matrix analysis, and routing observations. Section 6 discusses the findings and their implications. Section 7 summarizes limitations, and Section 8 concludes the paper and identifies future work.

1.9 Scope

The system focuses on binary text classification of SMS messages into HAM and SPAM. It does not independently verify sender identity, inspect the destination of a URL, detect malware on a device, or establish whether a message violates a particular law or policy. The reported confidence score is a model output and is not a guarantee of correctness.

2. RELATED WORK

2.1 Existing Approaches

SMS spam filtering inherits several ideas from email spam filtering, but the SMS environment has distinct characteristics. Messages are shorter, informal, and often contain abbreviations, telephone numbers, and compressed expressions. Delany, Buckley, and Greene reviewed SMS spam filtering methods and discussed data collection and benchmark challenges [1]. Their work highlights the importance of representative datasets and consistent evaluation.

Rule-based filtering uses predefined conditions to identify suspicious messages. Rules may search for specific expressions, promotional language, unusual character sequences, or known patterns. Such systems are easy to inspect, but they can be brittle when message wording changes. They can also flag legitimate messages containing the same terms.

Systematic reviews have examined the evolution of SMS spam detection and the use of machine-learning techniques [2], [3]. These reviews describe the transition from manually designed rules to statistical learning and more complex text-processing systems. The variety of datasets, preprocessing choices, feature representations, and evaluation metrics makes direct comparison across published studies difficult.

2.2 Machine-Learning Approaches

Traditional machine-learning methods learn a decision function from labelled examples. Naive Bayes estimates class probabilities from feature occurrences, while Logistic Regression estimates class membership using a learned linear function. Support Vector Machines identify a separating boundary between examples from different classes [4]–[6]. These methods are often computationally efficient and provide useful reference baselines.

For short text, the feature representation has a substantial effect on performance. TF-IDF weights terms according to their frequency within a message and their distribution across the corpus. The representation is commonly used with linear classifiers because it produces sparse vectors that can be processed efficiently [7], [8].

2. RELATED WORK (CONTINUED)

2.3 Deep-Learning Approaches

Deep-learning methods learn representations through multiple layers rather than relying only on manually specified features. Convolutional and recurrent architectures have been applied to text classification because they can capture local patterns and sequential relationships. CNN-based methods can identify local n-gram-like features, while recurrent networks model sequential dependencies.

Hybrid CNN–LSTM approaches have also been explored for SMS spam classification, including settings involving more than one language [9]. Other studies have investigated deep learning for filtering SMS spam and have reported strong results under their selected datasets and evaluation protocols [10]. These findings suggest that learned representations can be useful, although results remain dependent on data and experimental design.

2.4 Transformer-Based Approaches

The Transformer architecture uses self-attention to model relationships among tokens in a sequence [11]. BERT introduced bidirectional pre-training for language understanding and became a widely used foundation for downstream text-classification tasks [12]. DistilBERT applies knowledge distillation to create a smaller Transformer model while retaining substantial language-modeling capability [13].

Transformer-based methods have been investigated for SMS spam and phishing detection. A BERT-based approach to social-engineering and phishing-text detection reported high classification performance in its evaluated setting [14]. A later study explored bidirectional Transformer representations for short SMS messages and extended a benchmark dataset into more fine-grained spam categories [15].

ExplainableDetector investigated Transformer-based language modelling for SMS spam detection and incorporated explanation methods to provide additional information about model predictions [16]. A subsequent journal version appeared in Digital Communications and Networks [17]. These studies motivate further investigation into contextual text representations and analysis of model behaviour.

2. RELATED WORK (CONTINUED)

2.5 Hybrid and Multi-Feature Approaches

Hybrid methods combine more than one type of feature or representation. A multi-type feature extraction and early-fusion framework for SMS spam detection illustrates the use of complementary feature types in a combined model [18]. Other Transformer and ensemble-learning studies have also explored combining pretrained text representations with classifiers to improve spam-message classification [19].

Graph-enhanced Transformer methods provide another example of combining contextual representations with structural relationships. BERT-G3CN integrates BERT embeddings with multiple graph representations and reports results on two benchmark datasets [20]. Such approaches show that message content can be represented through more than one feature view.

2.6 Evasion, Robustness, and Explainability

Spam messages can be modified to evade detection. Spammers may alter spelling, insert punctuation, replace characters with digits, use abbreviations, or paraphrase known expressions. Research on evasive techniques in SMS spam filtering has examined the vulnerability of machine-learning models and anti-spam services to such modifications [21].

Explainability methods such as LIME and SHAP are used in machine learning to provide information about the features that influence predictions [22], [23]. These methods are relevant to text classification, but the presence of a routing weight or a confidence score should not automatically be treated as a complete explanation. Model interpretation requires separate validation.

2.7 Critical Comparison

Approach	Representation	Strength	Limitation
Rule-based filtering	Manually specified conditions	Easy to inspect	Requires continuous rule maintenance
TF-IDF + Logistic Regression	Sparse term weights	Efficient baseline	Limited contextual representation
TF-IDF + Linear SVM	Sparse term weights	Strong linear text classifier	Depends on feature construction
CNN / RNN models	Learned local or sequential features	Can capture patterns and sequence	Training and tuning required
Transformer models	Contextual token representations	Captures token relationships	Higher computational cost
Hybrid / graph models	Multiple feature views	Combines complementary signals	More architectural complexity
Proposed model	DistilBERT + three evidence pathways	Input-dependent evidence combination	Requires broader validation

2. RELATED WORK (CONTINUED)

2.8 Research Gap

Prior work has explored rule-based filtering, statistical text classification, deep-learning architectures, Transformer models, explainability methods, and multi-feature or graph-based approaches. These approaches demonstrate the value of learned text representations and complementary information sources.

The present study investigates a particular implementation that combines semantic, pattern, and context/entity evidence pathways using dynamic routing for SMS spam classification. The central research question is whether this combination produces useful classification performance on the selected test set and whether routing outputs provide descriptive information about model behaviour.

The experiments also reveal an important limitation: static equal fusion achieved the same aggregate classification metrics as the complete dynamic router on the reported test set. Therefore, the current results do not establish that dynamic routing is more accurate than static fusion. This observation defines a specific research gap for further controlled experimentation.

2.9 Summary of Literature Survey

The literature indicates that SMS spam detection has evolved from manually defined rules to statistical machine learning, deep learning, and contextual Transformer-based methods. TF-IDF-based classifiers offer efficient baselines, while Transformer models provide contextual text representations. Hybrid and graph-based systems demonstrate the potential of combining complementary representations.

The proposed project contributes an implementation and evaluation of a three-pathway evidence architecture with routing. Its novelty claim is limited to the specific model design and experimental study presented here; it does not claim to invent Transformer models, attention, or dynamic fusion as general techniques.

2.10 Comparison with Selected Published Studies

Study	Year	Reported result / contribution	Comparison caution
Deep learning to filter SMS spam [10]	2020	99.44% accuracy reported	Different setup; not head-to-head
Multi-Type Feature Extraction and Early Fusion [18]	2023	99.56% accuracy reported	Different feature design and evaluation
ExplainableDetector [17]	2025	RoBERTa reported at 99.84%	Different model and experimental protocol
BERT-G3CN [20]	2025	99.28% on UCI_SMS; 93.78% on ExAIS_SMS	Two benchmark datasets
Proposed Dynamic Evidence Routing	Current work	99.42% on reported 517-message test set	Evaluated against project baselines

These figures provide context only. The studies may use different dataset versions, train/test partitions, preprocessing, class distributions, and evaluation procedures. They must not be interpreted as a controlled ranking. A stronger comparison would reproduce the baselines using the same data split and preprocessing pipeline.

3. PROPOSED METHODOLOGY

3.1 Overall Framework

The proposed framework processes an SMS message through input preparation, Transformer representation, evidence extraction, dynamic routing, fusion, and final classification. The trained model is used during inference, while offline evaluation is performed separately using labelled test data.

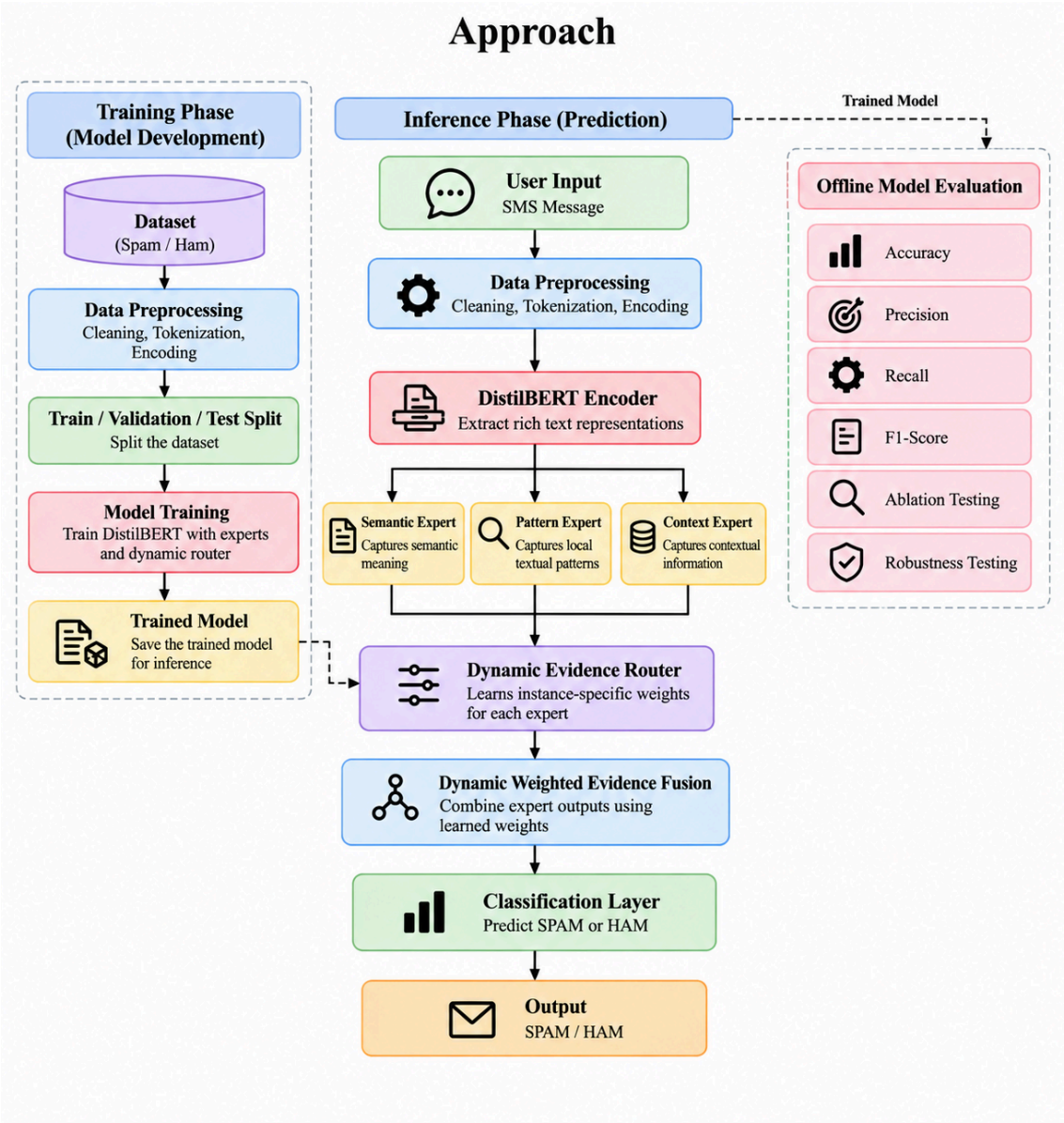


Fig. 1. Overall training, inference, and offline evaluation workflow.

The architecture contains a model-development phase and an inference phase. During model development, labelled messages are prepared and used to train the model. The trained checkpoint is saved for later inference. During inference, a user supplies a message, which is tokenized and passed through the DistilBERT encoder and evidence pathways. The classification output is returned to the user.

Offline evaluation is a separate process. It uses a labelled test set to calculate classification metrics, examine the confusion matrix, compare baselines, and perform ablation or robustness experiments. Evaluation metrics should not be represented as a step required for every individual prediction.

3.2 Dataset

The task is binary SMS classification. Each message belongs to one of two categories: legitimate messages, referred to as HAM, and unwanted or potentially fraudulent messages, referred to as SPAM. The project context identifies the SMS Spam Collection as the dataset family used for this work. The reported test set contains 517 messages.

Parameter	Description
Task	Binary SMS classification
Classes	HAM and SPAM
Test samples	517
Model	DistilBERT-based Dynamic Evidence Routing
Evaluation	Classification metrics, confusion matrix, baselines, ablation

3. PROPOSED METHODOLOGY (CONTINUED)

3.3 Data Preprocessing

Before classification, each SMS message must be converted into the representation expected by the Transformer model. The tokenizer maps the input text into token identifiers. Because a word may be divided into subword units, the token sequence can represent vocabulary items that are not present as complete words in the model vocabulary.

An attention mask is generated to distinguish meaningful token positions from padding positions. The token identifiers and attention mask are passed to the model. The same tokenizer and compatible preprocessing configuration must be used during training and inference. Inconsistent input preparation can cause prediction errors even when the saved checkpoint is valid.

The source paper does not provide a complete list of preprocessing operations, sequence length, truncation configuration, or normalization rules. These details should be added from the actual training and inference code. They should not be inferred from common Transformer defaults.

3.4 Feature Engineering

The proposed system uses a Transformer-based representation rather than relying exclusively on manually constructed term-frequency features. The DistilBERT representation is used as the basis for three evidence-processing pathways. Each pathway is intended to capture a different aspect of the message.

The Semantic Expert is intended to represent contextual meaning. The Pattern Expert is intended to identify message-level indicators and recurring textual patterns. The Context/Entity Expert is intended to represent contextual or entity-related evidence. These pathway descriptions explain their intended roles; the exact internal layer operations should be described from the implemented model code.

3.5 Feature Selection

The available project description does not identify a separate statistical feature-selection algorithm such as chi-square ranking, information gain, or recursive feature elimination for the proposed Transformer model. The report therefore does not claim that a separate feature-selection stage was implemented. In the proposed architecture, learned representations are processed by the evidence pathways and combined through routing.

3.6 Proposed Model

The proposed model is referred to in the implementation as DynamicEvidenceSMSModel. It combines a DistilBERT-based representation with three evidence pathways and a routing mechanism. The model produces a final binary classification and exposes routing information for analysis.

3. PROPOSED METHODOLOGY (CONTINUED)

3.7 DistilBERT Encoder

DistilBERT is a compact Transformer language model developed using knowledge distillation. The encoder processes a token sequence and generates contextual representations. Unlike a simple bag-of-words representation, contextual embeddings can encode relationships between tokens based on their surrounding sequence [12], [13].

Let an input message be represented by a token sequence X = (x1, x2, ..., xT), where T is the sequence length. Let A denote the corresponding attention mask. The encoder can be represented abstractly as H = DistilBERT(X, A), where H denotes the resulting hidden representation. This notation describes the model interface and does not specify undocumented layer-level implementation details.

3.8 Semantic Evidence Pathway

The Semantic Expert is intended to capture information associated with the meaning of the SMS message. Meaning often depends on relationships among words rather than the presence of an isolated expression. A message claiming that a recipient has won an unexpected reward and requesting an action to claim it contains a combination of signals that may be more informative than any single word.

The semantic pathway output may be represented as $z_s = f_s(H)$, where H is the encoder representation and f_s denotes the semantic processing component. The resulting representation is one source of evidence for final classification. The symbol is a conceptual description; the exact function must follow the implementation.

3.9 Pattern Evidence Pathway

The Pattern Expert is designed to capture information associated with recurring lexical or structural characteristics. SMS spam messages may share common expressions or patterns, while legitimate messages may contain different combinations of words and sentence structures.

The pattern pathway output may be represented as $z_p = f_p(H)$. Pattern evidence can be useful when a message contains recognizable signals such as promotional wording, urgency-related expressions, or requests to claim an offer. However, the same expression can appear in both HAM and SPAM messages, so no individual term should be treated as definitive evidence.

3. PROPOSED METHODOLOGY (CONTINUED)

3.10 Context/Entity Evidence Pathway

The Context/Entity Expert provides an additional source of evidence related to contextual or entity-associated information. Such information may help distinguish the significance of a phrase according to the surrounding message content. The presence of an organization name, telephone number, or other entity does not independently establish that a message is malicious because legitimate service notifications can contain these elements.

The context/entity pathway output may be represented as $z_c = f_c(H)$, where f_c denotes the implemented component. This representation is used as an additional input to the routing mechanism.

3.11 Dynamic Evidence Routing

The routing mechanism determines how the pathway representations contribute to the combined representation. Let the routing weights be α_s, α_p, and α_c for semantic, pattern, and context/entity evidence respectively. For a normalized weighted combination, the weights are non-negative and sum to one.

The combined representation can be written conceptually as $z = \alpha_s z_s + \alpha_p z_p + \alpha_c z_c$. This equation describes normalized weighted fusion. It should be treated as a mathematical abstraction of the model's intended combination; the exact operations and tensor dimensions should be verified against the implementation.

Dynamic routing means that relative contributions can vary between messages. The routing weights describe the model's internal combination and should not automatically be interpreted as calibrated probabilities, causal importance scores, or complete explanations of the final decision.

3.12 Evidence Fusion and Classification

After the pathway representations are combined, the classification component produces scores for HAM and SPAM. Let z denote the fused representation and g denote the classification function. The output scores can be represented as $o = g(z)$. If a softmax output layer is used, the class probabilities are calculated by normalizing the output scores. The predicted class is selected using the model's decision rule.

The system may display a confidence score associated with the prediction. A confidence score describes the model output; it does not guarantee correctness, particularly when an input differs from the examples represented during training.

3. PROPOSED METHODOLOGY (CONTINUED)

3.13 Training Objective

Supervised classification requires an objective that measures the difference between predicted outputs and target labels. Cross-entropy is a common objective for classification models. For a two-class problem, the standard expression is shown below. The exact objective used in this project should be confirmed against the training script before this expression is described as the implemented loss.

$$L_i = -\sum_{k=1}^{2} y_{i,k} \log(p_{i,k})$$

Here, $y_{i,k}$ denotes the target indicator for class k and $p_{i,k}$ denotes the predicted probability for that class. For N training examples, the average loss is:

$$L = -(1/N) \sum_{i=1}^{N} \sum_{k=1}^{2} y_{i,k} \log(p_{i,k})$$

During training, model parameters are adjusted to reduce the selected loss. The trained parameters are saved in a checkpoint that can later be loaded for inference. The optimizer, learning rate, batch size, number of epochs, and checkpoint-selection procedure must be taken from the actual training configuration. These values are not specified in the source material available for this report.

3.14 Inference Procedure

During inference, the trained model receives a new SMS message and applies the preprocessing and tokenization required by the model. The resulting token identifiers and attention mask are passed through the forward computation. The Transformer produces a representation, the evidence pathways generate their representations, and the routing mechanism combines the available evidence. The classification component then generates the output used to predict HAM or SPAM.

The implementation can return the predicted label and available confidence information. If routing values are exposed, they can be recorded for later analysis. This allows the researcher to examine both the final classification and the relative contribution of the evidence pathways.

3. PROPOSED METHODOLOGY (CONTINUED)

3.15 Algorithmic Description

Algorithm 1 summarizes the functional workflow of the Dynamic Evidence Routing SMS classifier.

- Input: SMS text x and trained model parameters θ.
- Output: Predicted class, confidence information, and available routing values.
- Receive the SMS text.
- Tokenize the message using the model-associated tokenizer and generate the attention mask.
- Compute the Transformer representation using DistilBERT.
- Generate semantic, pattern, and context/entity pathway representations.
- Determine routing weights using the implemented routing mechanism.
- Combine pathway representations according to the implemented fusion operation.
- Compute classification scores and apply the model's decision rule.
- Return the predicted class and available confidence and routing information.

3.16 Methodological Considerations

The proposed architecture introduces three main evaluation considerations. First, each pathway should provide useful information beyond the other pathways. Second, dynamic routing should be compared with a fixed combination strategy. Third, the model should be tested on messages that differ from training examples.

The ablation study addresses the first two considerations by examining variants with individual pathways removed and a static equal-fusion configuration. The baseline comparison evaluates the complete system against conventional TF-IDF classifiers. These experiments provide evidence about performance under the reported evaluation conditions, but they do not establish that routing will improve every dataset.

3.17 Reproducibility Requirements

For reproducibility, a final implementation record should include the exact dataset version and source, class distribution, split proportions, random seed, tokenizer name and revision, maximum sequence length, optimizer, learning rate, batch size, epoch count, loss function, hardware, software versions, and checkpoint-selection procedure. Where these details are unavailable in the source material, they are explicitly left for confirmation rather than fabricated.

4. EXPERIMENTAL SETUP

4.1 Implementation Environment

The proposed SMS spam detection system was implemented using Python and deep-learning libraries. The model uses a DistilBERT-based text representation component together with multiple evidence-processing pathways and a dynamic evidence-routing mechanism. The implementation is organized into separate components for model definition, training, evaluation, and application-level prediction.

PyTorch is used for model implementation and tensor operations, while the Transformers library provides the tokenizer and Transformer-related functionality. The trained model parameters are stored in a checkpoint file and loaded during inference. A Flask-based web application provides an interface through which users can enter an SMS message and receive a predicted class.

Software / Library	Role in the project
Python	Programming language
PyTorch	Model implementation and tensor operations
Hugging Face Transformers	DistilBERT and tokenizer functionality
scikit-learn	TF-IDF baseline classifiers and evaluation utilities
Flask	Web application and inference interface
pandas / NumPy	Data handling and numerical operations

4.2 Dataset and Evaluation Split

The reported test set contains 517 SMS messages. The same reported test set was used for the proposed model and the two traditional machine-learning baselines. Using a common test set makes the comparison more informative because the models are evaluated on the same collection of messages.

The source document does not specify the exact training, validation, and test proportions, the random seed, or the detailed class distribution of the complete dataset. These details must be extracted from the actual project configuration before submission. The test-set count of 517 is retained as reported.

4.3 Proposed Model Configuration

The model evaluated in this study is the saved-checkpoint version of DynamicEvidenceSMSModel. It combines DistilBERT-based text representations with semantic, pattern, and context/entity pathways. The dynamic evidence-routing mechanism combines information from these pathways to produce the final classification.

4. EXPERIMENTAL SETUP (CONTINUED)

4.4 Baseline Models

Two conventional machine-learning baselines were evaluated alongside the proposed model. The first baseline combines Term Frequency–Inverse Document Frequency (TF-IDF) text representation with Logistic Regression. TF-IDF represents messages according to the importance of their terms, while Logistic Regression estimates class membership from the resulting feature vectors.

The second baseline combines TF-IDF with a Linear Support Vector Machine. The classifier learns a separating boundary between legitimate and spam messages using their vector representations. Linear SVM is a useful comparison because it can perform well in high-dimensional text-classification tasks without requiring a Transformer-based architecture.

Baseline	Representation	Classifier
Baseline 1	TF-IDF	Logistic Regression
Baseline 2	TF-IDF	Linear Support Vector Machine
Proposed	DistilBERT and three evidence pathways	Dynamic Evidence Routing classifier

4.5 Experimental Procedure

- Load the saved model checkpoint and compatible tokenizer.
- Prepare the labelled test messages using the model's input-processing procedure.
- Generate predictions for the test messages.
- Compare predicted labels with the reference labels.
- Calculate accuracy, precision, recall, and F1-score.
- Generate the confusion matrix and inspect error counts.
- Evaluate the two TF-IDF baselines on the same reported test set.
- Run the reported ablation configurations and record their metrics.
- Analyze average routing weights and representative message-level routing values.

4.6 Cross-Validation and Statistical Testing

The available project results report a single test-set evaluation and do not provide repeated cross-validation results, confidence intervals, or statistical significance tests. This paper therefore does not claim that the reported differences are statistically significant. Repeated runs and suitable paired statistical tests are recommended for future validation.

4.7 Hyperparameter Configuration

The source material does not provide verified values for the optimizer, learning rate, batch size, number of epochs, maximum sequence length, weight decay, or random seed. These settings must be copied from the training configuration. Reporting guessed values would reduce reproducibility and is intentionally avoided.

4. EXPERIMENTAL SETUP (CONTINUED)

4.8 Flask-Based Application

The trained model is connected to a Flask-based web application. The application provides a message-entry interface through which a user can submit SMS text for prediction. The submitted message is processed using the saved model checkpoint and compatible tokenizer.

The application returns a predicted category, either HAM or SPAM, together with a confidence score and routing information where available. Separating the model's training and inference components allows the saved checkpoint to be evaluated independently of the web interface.

The Flask interface demonstrates a prototype workflow for accessing the trained classifier. It does not by itself establish production readiness. Before deployment, the application would require security testing, input validation, performance measurements, access controls, and privacy safeguards.

4.9 Application Workflow

- The user opens the web application.
- The user enters an SMS message in the input field.
- The application passes the message to the inference component.
- The tokenizer converts the message into model inputs.
- The trained model generates a classification output.
- The application displays the predicted class and available confidence and routing values.

4.10 Hardware and Software Reporting

The project documentation identifies Python, PyTorch, Transformers, scikit-learn, and Flask as implementation technologies. It does not provide a verified hardware specification or exact library versions. The final reproducibility record should state the operating system, processor, GPU model and memory if used, Python version, package versions, and inference environment.

The application should be evaluated separately from the offline model metrics. Interface functionality demonstrates that the model can be accessed through a web workflow, while model performance must be established through labelled test data.

5. RESULTS

5.1 Overall Performance

The proposed Dynamic Evidence Routing model was evaluated on a test set containing 517 SMS messages. The reported results are 99.42% accuracy, 100.00% precision, 95.38% recall, and 97.64% F1-score. These values describe the model's performance on the reported test set and should not be treated as guaranteed performance on other datasets.

Metric	Reported value	Interpretation
Accuracy	99.42%	Proportion of all test messages classified correctly
Precision	100.00%	No false-positive spam predictions in this test set
Recall	95.38%	Most actual spam messages were detected
F1-score	97.64%	Harmonic mean of precision and recall
Test messages	517	Number of evaluated messages

The precision of 100.00% indicates that every message predicted as SPAM in this test set was labelled SPAM. The recall of 95.38% indicates that three actual SPAM messages were classified as HAM. The F1-score balances precision and recall and provides a useful summary when both types of error matter.

5.2 Confusion Matrix

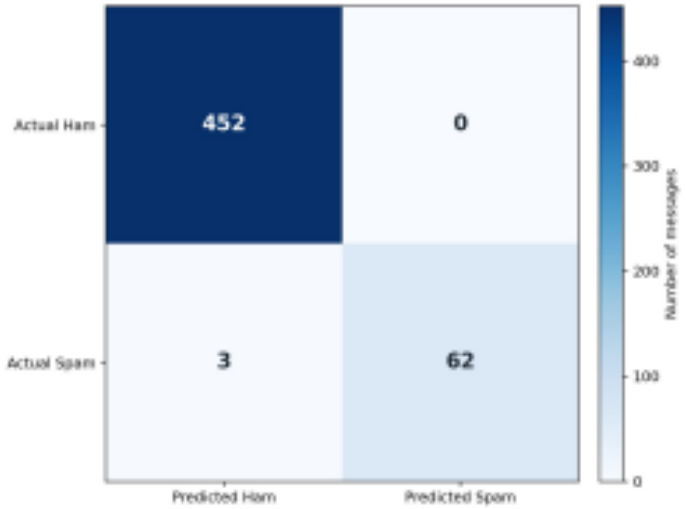


Fig. 2. Confusion matrix of the proposed model.

The confusion matrix contains 452 true negatives, zero false positives, three false negatives, and 62 true positives. The absence of false positives means that no legitimate message in the reported test set was incorrectly classified as spam. The three false negatives indicate that the classifier did not detect every spam message.

Actual class	Predicted HAM	Predicted SPAM
HAM	452	0
SPAM	3	62

5. RESULTS (CONTINUED)

5.3 Comparison with Baseline Models

The proposed model was compared with TF-IDF plus Logistic Regression and TF-IDF plus Linear SVM using the reported common test set. The source paper reports the following accuracy, recall, and F1-score values in its baseline comparison table and accompanying discussion. The Logistic Regression recall is inconsistent between the original table and narrative; the narrative value of 70.77% is used here because it is consistent with the stated precision of 100.00% and F1-score of 82.88%.

Model	Accuracy	Recall	F1-score
TF-IDF + Logistic Regression	96.32%	70.77%	82.88%
TF-IDF + Linear SVM	98.07%	87.69%	91.94%
Proposed Dynamic Evidence Routing	99.42%	95.38%	97.64%

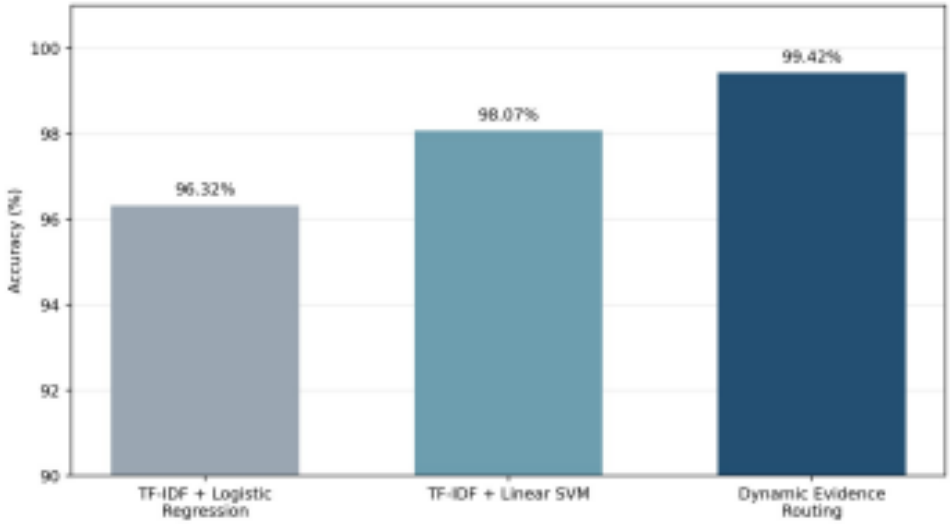


Fig. 3. Accuracy comparison of the evaluated models.

The Linear SVM baseline improved on Logistic Regression in accuracy, recall, and F1-score according to the reported values. The proposed model achieved the highest accuracy and F1-score among the three evaluated configurations. Compared with Linear SVM, its recall was higher by 7.69 percentage points and its F1-score was higher by 5.70 percentage points.

These results support the conclusion that the proposed model performed better than the two tested baselines on this evaluation. They do not establish that it outperforms all existing SMS spam detection methods. The comparison is limited to the two implemented baselines and the reported test set.

5. RESULTS (CONTINUED)

5.4 Comparison with Published Studies

Published SMS spam detection studies report a range of performance values. These results are useful for positioning the current work within the broader literature, but they are not direct head-to-head comparisons unless the same dataset, split, preprocessing, and evaluation procedure are used.

Published approach	Year	Reported accuracy
Deep learning to filter SMS spam [10]	2020	99.44%
Multi-Type Feature Extraction and Early Fusion [18]	2023	99.56%
ExplainableDetector / RoBERTa [17]	2025	99.84%
BERT with multi-graph convolutional networks [20]	2025	99.28% on UCI_SMS
Proposed Dynamic Evidence Routing	Current work	99.42%

The proposed model's reported accuracy is higher than the 99.28% reported for BERT-G3CN on the UCI_SMS dataset, but lower than the selected reported accuracies of 99.44%, 99.56%, and 99.84%. The underlying datasets and protocols differ, so this table is contextual rather than a controlled performance ranking.

5.5 False-Positive and False-Negative Analysis

False positives and false negatives have different practical consequences. A false positive can cause a legitimate message to be hidden or filtered. A false negative allows a spam message to remain in the user's inbox. The reported confusion matrix contains no false positives and three false negatives.

The exact reasons for the three missed spam messages cannot be established without examining their text and the model's intermediate outputs. Future analysis should inspect these examples to determine whether they contain unfamiliar wording, indirect intent, or patterns that differ from the training examples.

5.6 Summary of Results

The results indicate strong performance on the reported test set. The proposed model achieved 99.42% accuracy and a 97.64% F1-score. The baseline comparison suggests improved recall and F1-score relative to the two tested conventional models, subject to the documented correction of the Logistic Regression recall inconsistency.

5. RESULTS (CONTINUED)

5.7 Ablation Study

An ablation study evaluates how performance changes when individual components are removed or when the model's combination strategy is modified. It helps investigate whether the complete architecture provides measurable benefits over simplified configurations.

The proposed architecture includes the Semantic Expert, Pattern Expert, Context/Entity Expert, and Dynamic Evidence Routing mechanism. The ablation experiments compare the complete model against configurations that exclude one expert or replace dynamic routing with static equal fusion.

Configuration	Accuracy	Precision	Recall	F1-score
Full Dynamic Router	99.42%	100.00%	95.38%	97.64%
Without Semantic Expert	99.23%	100.00%	93.85%	96.83%
Without Pattern Expert	99.23%	98.41%	95.38%	96.88%
Without Context/Entity Expert	99.42%	100.00%	95.38%	97.64%
Static Equal Fusion	99.42%	100.00%	95.38%	97.64%

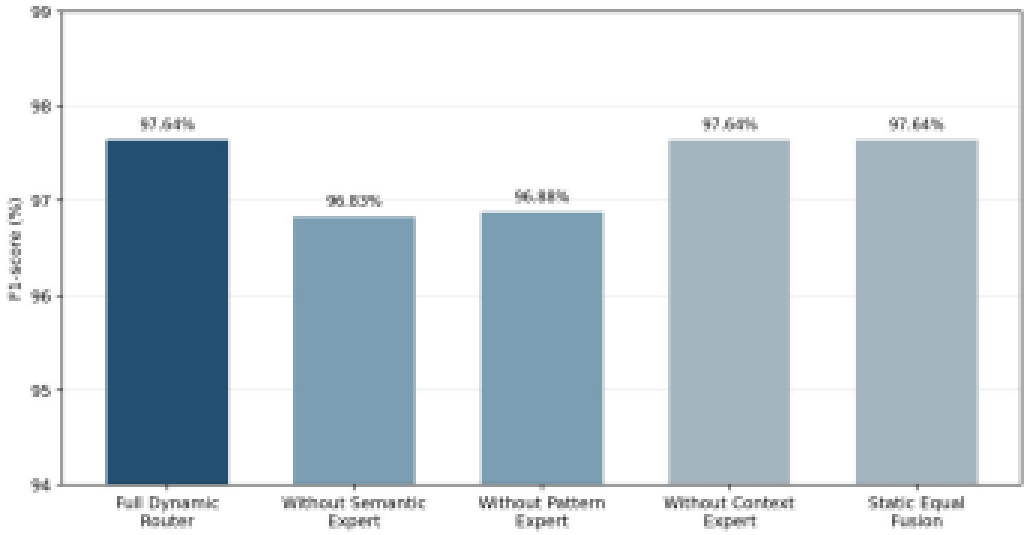


Fig. 4. F1-score comparison across ablation configurations.

The complete Dynamic Router achieved the highest F1-score among the configurations shown. Removing the Semantic Expert reduced recall from 95.38% to 93.85%, suggesting that semantic evidence contributed to detecting some spam messages in the evaluated set. Removing the Pattern Expert reduced precision from 100.00% to 98.41%, while recall remained at 95.38%.

Removing the Context/Entity Expert and using static equal fusion produced the same reported aggregate metrics as the complete model. Identical aggregate metrics do not prove that these components are unnecessary; individual predictions or internal representations may differ while aggregate scores remain unchanged.

5. RESULTS (CONTINUED)

5.8 Analysis of Individual Evidence Pathways

The Semantic Expert is intended to capture contextual meaning that may not be fully represented by individual keywords or simple message patterns. Removing it reduced recall by approximately 1.54 percentage points and reduced F1-score by approximately 0.81 percentage points. This suggests that semantic information was useful for the evaluated configuration, although the contribution may vary across messages.

The Pattern Expert is designed to identify message-level indicators associated with spam, such as suspicious phrases, promotional wording, urgency, or requests to claim an offer. Removing this pathway reduced precision from 100.00% to 98.41%, while recall remained unchanged. The result suggests that pattern evidence may have contributed to the reliability of spam predictions in this evaluation.

The Context/Entity Expert is intended to represent contextual or entity-related evidence relevant to message classification. Removing it did not change the reported accuracy, precision, recall, or F1-score. This is an observation about the current test set, not a general conclusion that contextual evidence is unnecessary.

5.9 Dynamic Routing Versus Static Equal Fusion

Dynamic evidence routing is intended to adapt the relative contribution of the evidence pathways to the input message. Static equal fusion combines the pathways using a fixed equal-weight strategy. The reported results show that static equal fusion achieved the same aggregate classification metrics as the full dynamic router.

Therefore, the current experiment does not demonstrate an improvement in aggregate classification performance from dynamic routing over static equal fusion on this test set. Further evaluation should compare prediction-level differences, routing distributions, computational costs, and performance across multiple test sets.

5.10 Average Routing Distribution

The model's reported average routing distribution was 14.38% for semantic evidence, 82.60% for pattern evidence, and 3.02% for context/entity evidence. The pattern pathway received the largest average share of routing weight across the evaluated messages.

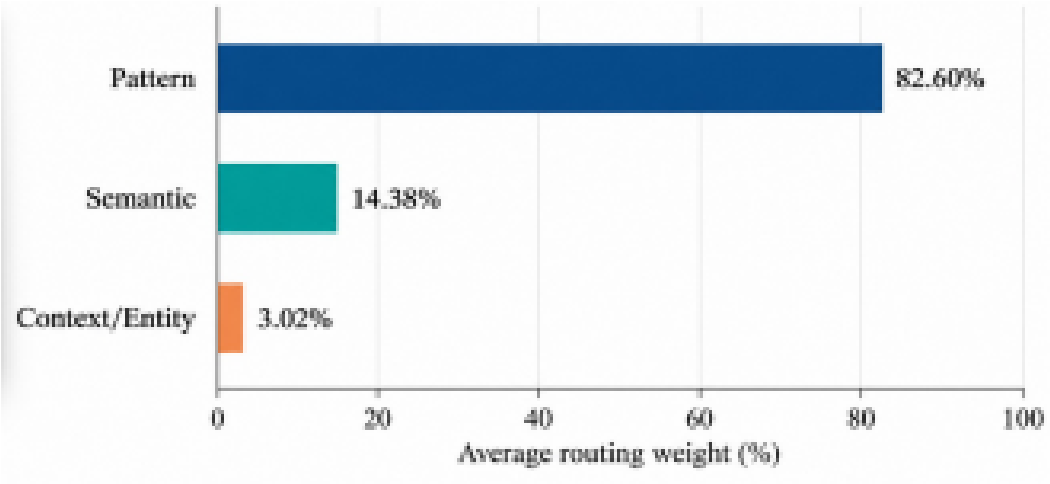


Fig. 5. Average evidence-routing distribution.

5. RESULTS (CONTINUED)

5.11 Interpretation of Routing Weights

The average distribution describes the model's reported routing behaviour. It should not be interpreted as proof that pattern evidence is inherently more important than semantic evidence. The weights depend on the learned model, dataset, architecture, and implementation. A smaller average weight also does not necessarily mean that a pathway has little influence on particular messages.

5.12 Message-Level Routing Behaviour

Message type	Semantic	Pattern	Context/Entity
Example spam message	74.34%	7.81%	17.85%
Example legitimate message	7.40%	89.52%	3.08%

The example spam message, “Congratulations! You have won a free prize. Call now to claim,” received a spam prediction with a reported confidence of 96.69%. Its routing distribution assigned 74.34% to semantic evidence, 7.81% to pattern evidence, and 17.85% to context/entity evidence.

The example legitimate message, “Hey, are we still meeting for lunch today?”, received a HAM prediction with a reported confidence of 98.99%. Its routing distribution assigned 7.40% to semantic evidence, 89.52% to pattern evidence, and 3.08% to context/entity evidence.

These examples illustrate that routing weights can differ across inputs. The spam example relied more heavily on semantic evidence, whereas the legitimate example assigned most of its routing weight to pattern evidence. Two messages are insufficient to establish a general rule about how the model routes all spam and legitimate messages.

5.13 Results Summary

The ablation and routing experiments provide two types of information. First, ablation results show how classification metrics changed when components were removed or the fusion strategy was modified. Second, routing distributions illustrate how the model allocated evidence for individual messages and across the evaluated set. Further evaluation is needed to establish whether dynamic routing provides consistent advantages over simpler fusion methods.

6. DISCUSSION

6.1 Interpretation of Findings

The experimental results indicate that the proposed Dynamic Evidence Routing model performed strongly on the reported SMS spam detection test set. Its accuracy of 99.42% and F1-score of 97.64% demonstrate that it correctly classified most of the evaluated messages. The confusion matrix shows that it produced no false-positive predictions and missed three actual spam messages.

The baseline comparison provides additional context. TF-IDF with Logistic Regression achieved lower reported recall, while TF-IDF with Linear SVM achieved a recall of 87.69%. The proposed model achieved a recall of 95.38%. This difference suggests that the proposed configuration detected a larger proportion of spam messages in the same reported test set.

The results are consistent with the possibility that combining Transformer representations with multiple evidence pathways can support SMS classification. However, only two conventional baselines were evaluated. The findings should therefore be restricted to these specific comparisons and should not be interpreted as universal superiority over all existing models.

6.2 Importance of Evidence Combination

SMS messages vary considerably in length, vocabulary, and intent. Some unwanted messages contain explicit promotional language, while others use indirect expressions. Legitimate messages can also contain words commonly associated with spam, depending on the surrounding message. Combining semantic, pattern, and context/entity pathways provides a framework for representing different aspects of a message.

The ablation results suggest that the Semantic and Pattern pathways affected selected metrics in the reported evaluation. The context/entity pathway did not change aggregate scores when removed. These observations should be investigated through repeated experiments and message-level analysis.

6.3 Dynamic Routing and Its Interpretation

The routing outputs show that the relative weights differed between the two example messages. The average routing distribution was dominated by pattern evidence, while the spam example assigned the largest weight to semantic evidence. These observations illustrate input-dependent behaviour, but they do not establish that routing weights are complete explanations or causal measures of component importance.

6. DISCUSSION (CONTINUED)

6.4 Comparison with Previous Studies

Prior studies have explored classical machine-learning models, deep-learning architectures, Transformer-based classifiers, graph-enhanced representations, and explainability methods for SMS spam detection [1]–[21]. Several published studies report high accuracy values, including results above 99% on selected datasets.

The proposed model achieved 99.42% accuracy on its reported test set. This value is close to several published results, but differences in datasets and evaluation protocols prevent a direct ranking. In particular, a high score on a small test set does not necessarily indicate better generalization than a model evaluated on a larger or more diverse dataset.

6.5 Theoretical Implications

The project provides an implementation-level examination of evidence combination in short-text classification. It illustrates how a contextual Transformer representation can be processed by separate pathways and combined using a routing mechanism. The ablation results also demonstrate why component-level evaluation is necessary: the presence of an additional module does not guarantee a measurable change in aggregate performance.

The equality of metrics between dynamic routing and static equal fusion is an important result. It suggests that future work should investigate whether routing provides benefits beyond the final aggregate scores, such as different message-level decisions, more stable performance across datasets, or better robustness.

6.6 Practical Implications

The proposed system may be useful as a research prototype for SMS spam filtering. Its classification output can help distinguish likely spam from legitimate messages, while confidence scores and routing information can provide additional information about individual predictions. The Flask interface demonstrates a possible way to connect the trained classifier to a user-facing application.

A working interface is not equivalent to a production-ready spam filtering service. Before deployment, the system would require testing under realistic message traffic, monitoring of false-positive and false-negative errors, security checks, performance testing, and procedures for updating the model as spam behaviour changes.

6.7 Unexpected Findings

Two ablation configurations produced the same aggregate metrics as the complete model: removal of the context/entity expert and replacement of dynamic routing with static equal fusion. This finding limits the strength of any claim that each component is essential for accuracy on the current test set. It also motivates further controlled analysis.

7. LIMITATIONS

7.1 Dataset Limitations

The reported evaluation uses a test set of 517 messages. Although the results are encouraging, a test set of this size may not represent the full range of messages encountered in practice. Spam campaigns, language use, message formats, and legitimate communication styles can vary across regions, users, and time periods.

The dataset source and detailed composition should be explicitly documented in the final reproducibility record. The project context identifies the SMS Spam Collection dataset family, but the exact dataset version, class distribution, and complete splitting procedure should be confirmed from the files used during training.

7.2 Evaluation Limitations

The reported metrics describe performance on one test set and should not be treated as guaranteed performance on unseen datasets. The observed precision of 100.00% means that no false positives occurred in this evaluation; it does not prove that false positives will never occur. Similarly, the model's 95.38% recall indicates that some spam messages were missed.

Repeated experiments using different random seeds, additional evaluation datasets, and confidence intervals would help assess the stability of the results. Statistical significance testing would also help determine whether differences between the proposed model and baselines are larger than would reasonably be expected from sampling variation.

7.3 Ablation Limitations

The ablation study shows that removing the Semantic Expert or Pattern Expert changed some classification metrics. However, removing the Context/Entity Expert and replacing dynamic routing with static equal fusion did not change the reported aggregate scores. These observations do not establish that the affected components are universally useful or unnecessary.

Future ablation studies should compare predictions at the message level, repeat training under controlled conditions, and test configurations on more diverse datasets. This would provide a more complete understanding of the contribution of each component.

7.4 Robustness and Generalization

Robustness is important because spam messages can change their wording, introduce unusual spelling, use abbreviations, or attempt to avoid familiar patterns. A preliminary robustness test involving only a small number of examples is insufficient to support a broad claim of robustness. A more reliable evaluation should include larger collections of deliberately varied messages, including spelling changes, paraphrases, unseen promotional phrases, and legitimate messages containing suspicious-looking words.

7. LIMITATIONS (CONTINUED)

7.5 Computational and Deployment Limitations

The proposed model uses a Transformer-based architecture, which generally requires more computational resources than a simple TF-IDF classifier. The actual cost depends on the model configuration, sequence length, hardware, batch size, and inference implementation.

Although the application can produce predictions through Flask, further measurements are needed before making claims about speed or scalability. Relevant measurements include average inference latency, memory consumption, throughput, and resource usage under concurrent requests.

A practical deployment should protect message content, restrict access to model files, validate incoming requests, and avoid retaining sensitive SMS content unnecessarily. These considerations are important when handling personal communication.

7.6 Explainability Limitations

The routing weights indicate how the model allocates relative contributions among its evidence pathways. They do not necessarily identify the exact words responsible for a prediction, prove causal importance, or provide a complete explanation of the model's decision. Explanation methods should be evaluated separately if the system is intended to support high-stakes filtering decisions.

7.7 Limitations Summary

The main limitations of the current work are the relatively small reported test set, the need to document the exact dataset and training configuration, the limited number of evaluated baselines, and the absence of sufficiently large robustness testing. In addition, the ablation results do not show a measurable aggregate improvement from dynamic routing over static equal fusion on the current test set.

These limitations do not invalidate the reported results. Rather, they define the scope of the conclusions and identify the experiments needed to strengthen the evidence for the proposed method.

8. CONCLUSION AND FUTURE WORK

8.1 Conclusion

This research presented an SMS spam detection approach based on a DistilBERT-based model and dynamic evidence routing. The proposed architecture combines semantic, pattern, and context/entity evidence pathways to support binary classification of SMS messages into spam and ham. A Flask-based interface was also used to demonstrate how the trained model can be accessed through a message-entry application.

On the reported test set of 517 messages, the proposed model achieved 99.42% accuracy, 100.00% precision, 95.38% recall, and a 97.64% F1-score. Its confusion matrix contained 452 true negatives, 62 true positives, three false negatives, and no false positives.

The model was compared with TF-IDF plus Logistic Regression and TF-IDF plus Linear SVM. The proposed model achieved higher reported recall and F1-score than both baselines on the same test set. These results suggest that the implemented architecture is promising for SMS spam classification under the evaluated conditions.

The ablation study provided additional information about the model's components. Removing the Semantic Expert reduced recall, while removing the Pattern Expert reduced precision. Removing the Context/Entity Expert and using static equal fusion produced the same aggregate metrics as the complete model. Therefore, the current experiments support the effectiveness of the overall classifier on the test set, but they do not establish that dynamic routing consistently outperforms simpler fusion strategies.

The research provides a basis for further investigation into evidence combination for SMS spam detection. More extensive evaluation is needed to establish generalization, robustness, computational efficiency, and practical reliability.

8.2 Future Work

- Evaluate the model on larger and more diverse SMS datasets from different sources and time periods.
- Compare the proposed model with a properly trained DistilBERT-only classifier.
- Repeat experiments using multiple random seeds and report confidence intervals.
- Compare dynamic routing and static fusion under controlled conditions.
- Analyze the three false-negative predictions to understand why those spam messages were missed.
- Conduct larger robustness tests using spelling variations, abbreviations, paraphrases, and unusual formatting.
- Measure inference latency, memory usage, and throughput under realistic workloads.
- Strengthen application security, privacy controls, and monitoring before any real-world deployment.

DECLARATIONS

Funding

The authors report that no specific external funding information was provided in the project materials. If the project received institutional, grant, or sponsor support, the funding body and grant details should be inserted before submission.

Conflict of Interest

The authors should declare any financial or personal relationships that could influence the work. No conflict-of-interest statement was supplied in the source project material; the authors should confirm the appropriate declaration with their supervisor.

Ethics Statement

The study uses labelled SMS text for machine-learning research. The project materials do not provide a formal ethics approval identifier or details of participant recruitment. If the dataset is a public, de-identified benchmark, state the verified source and applicable terms of use. Do not claim ethics approval unless it was obtained.

Data Availability

The project context identifies the SMS Spam Collection dataset family. The exact version, source URL, and any preprocessing or derived files used in the experiments should be specified by the authors. The trained model checkpoint and code availability should be described according to the actual sharing arrangements.

Author Contributions

Author contribution roles were not individually specified in the source document. The authors should confirm and insert the contributions for conceptualization, methodology, software, validation, formal analysis, investigation, writing, and supervision.

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The references below cover SMS spam filtering, text classification, Transformer models, evaluation, and implementation tools. Bibliographic details should be checked against the publisher records before formal journal submission.

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AUTHOR CHECKLIST BEFORE SUBMISSION

This final page records details that must be verified against the original notebook, model code, and dataset files before the paper is submitted. It is included to prevent unverified settings or inconsistent metrics from being presented as established results.

Item to verify	Current status / action
Dataset source and exact version	Confirm the exact dataset file, source URL, and version used.
Train/validation/test split	Insert the exact split proportions and sample counts from the notebook.
Random seed	Insert the seed used for splitting and model training, if set.
Preprocessing and tokenizer	Record the exact tokenizer, maximum sequence length, truncation, and padding settings.
Training configuration	Confirm optimizer, learning rate, batch size, epochs, loss function, and checkpoint selection.
Hardware and package versions	Insert verified operating system, CPU/GPU, Python, PyTorch, Transformers, and scikit-learn versions.
Baseline Logistic Regression	The source table lists 100.00% recall, while the narrative reports 70.77%. This paper uses 70.77%, consistent with the reported 100.00% precision and 82.88% F1; verify from notebook.
Proposed-model recall	Use 95.38%, consistent with the confusion matrix (62/65 spam detected) and ablation table; remove any 100.00% recall entry for the proposed model.
Statistical testing	No repeated-run confidence intervals or significance tests were supplied. Do not claim statistical significance.
Robustness testing	Do not claim broad robustness from a very small illustrative test.
Declarations	Confirm funding, conflict of interest, ethics, data availability, and individual author contributions.

The document follows the supplied research-paper structure: Abstract, Introduction, Related Work, Proposed Methodology, Experimental Setup, Results, Discussion, Limitations, Conclusion and Future Work, Declarations, Acknowledgements, and References. The reference guide requests at least 30 references; this version includes 30 entries. Bibliographic details should be checked against publisher records before formal submission.